

Early Cancer Prediction - Project Report

1. Project Overview

The 'Early Cancer Prediction' project is an intelligent healthcare application designed to detect three prominent types of cancer: Brain Tumors, Oral Cancer, and Cervical Cancer at their early stages. The system leverages Machine Learning and Deep Learning models developed using comprehensive medical datasets.

2. How the Application is Running

The application relies on a robust architecture to serve predictions:

Backend Framework: The backend is built on a Flask web server (app.py). It acts as the bridge between the user interface and the machine learning models.

Model Loading: Pre-trained models (serialized using joblib) are stored in the 'Models/' directory. These models are loaded into memory when the server starts up, allowing for fast, real-time predictions.

RESTful APIs: The server exposes several API endpoints such as:

- /api/predict/brain-tumor
- /api/predict/oral-cancer
- /api/predict/cervical-cancer

Preprocessing & Validation: For image-based models (Brain and Oral), the app includes image validation logic to ensure users upload the correct type of image (e.g., verifying grayscale MRIs versus color photos). Images are then preprocessed (resized to 64x64, padded, and normalized) before being fed to the models. Cervical cancer data uses a pre-fitted StandardScaler.

3. Model Specifications & Details

Below is a breakdown of the models, their underlying algorithms, and their performance metrics:

A. Brain Tumor Prediction Model

- Algorithm: Random Forest Ensemble utilizing CNN-extracted features.
- Input Data: Grayscale Brain MRI Scans (preprocessed to 64x64 array).
- Classes Detected: Glioma, Meningioma, Pituitary, and No Tumor (Healthy).
- Performance Metrics: Accuracy - 95.8%, F1 Score - 0.957, Precision - 0.960.
- Key Detail: The application actively validates uploaded images to ensure they meet the color and contrast profiles of an actual Brain MRI, preventing false readings from standard photos.

B. Oral Cancer Prediction Model

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- Algorithm: Ensemble Voting Mechanism combining Random Forest, Support Vector Machine (SVM), and Decision Tree classifiers.
- Input Data: Clinical images of oral mucosa and lesions.
- Classes Detected: Cancerous, Non-Cancerous.
- Performance Metrics: Accuracy - 93.4%, F1 Score - 0.931, Precision - 0.940.
- Key Detail: The application combines predictions from all three models using a majority voting system to determine the final prediction, ensuring high robustness.

C. Cervical Cancer Prediction Model

- Algorithm: Soft Voting Ensemble (combining Random Forest, AdaBoost, SVM, K-Nearest Neighbors, and Decision Tree).
- Input Data: 33 Tabular risk factors including Age, Number of sexual partners, Smoking habits, and STD/HPV diagnosis history.
- Performance Metrics: Accuracy - 96.5%, F1 Score - 0.962, Precision - 0.970.
- Key Detail: Alongside the ML model's prediction, the app calculates a 'Clinical Score' based on weighted high-risk factors. These scores are blended to provide a comprehensive Risk Percentage to the patient.